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A Reproducible ML Pipeline for Ovarian Cancer Platinum-Response Prediction with Empirical Quantification of Feature-Selection Leakage

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Software

- Review
- Repository
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Summary

`ovarian-cancer-platinum-response` is a modular, reproducible Python pipeline for predicting platinum chemotherapy response in ovarian cancer patients using bulk RNA-seq gene expression profiles from the TCGA Ovarian Serous Cystadenocarcinoma cohort (The Cancer Genome Atlas Research Network, 2011). The software trains and evaluates three supervised classifiers — Logistic Regression, Random Forest, and Support Vector Machine — across three survival thresholds (12, 18, and 24 months), and provides evaluation tools including cross-validated ROC-AUC with bootstrap confidence intervals, SHAP explainability (Lundberg & Lee, 2017), and Kaplan-Meier survival stratification (Davidson-Pilon, 2019).

Beyond the core prediction pipeline, the package includes a paired ablation study that empirically quantifies AUC inflation from a widely documented but poorly specified failure mode in genomic machine learning: computing feature selection statistics on the full dataset before the train/test split. The key contribution is the distinction between unsupervised variance-based filtering (mean inflation: +0.01 AUC, non-significant across 30 repeated splits) and supervised F-score-based selection (mean inflation: +0.22 AUC, up to +0.55 for SVM, significant at $p<0.0001$ in 7 of 9 model/threshold combinations). A label construct-validity investigation further examines whether the OS-threshold proxy labels standard in this literature capture the clinical construct they are assumed to represent.

Statement of Need

Machine learning applied to cancer genomics regularly reports optimistic performance, a problem attributed in part to feature selection leakage (Teschendorff, 2020; Vabalas et al., 2019). However, published guidance rarely distinguishes between supervised selection methods — which directly use outcome labels, creating a direct information pathway from test fold to training — and unsupervised methods, which use only expression variance and create only an indirect pathway. This distinction matters empirically: our ablation shows the two conditions produce dramatically different inflation magnitudes (mean +0.22 vs. +0.01) on identical data and models. Existing genomic ML tools (Davis & Meltzer, 2007; Zararsiz et al., 2018) do not include reproducible ablation frameworks for quantifying this effect with paired statistical testing across repeated splits.

A second unmet need is tooling to investigate label construct validity. TCGA-OV lacks a direct platinum-response label; researchers use OS-threshold surrogates without empirically verifying how well they capture the intended clinical construct. The software's `label_investigation` module provides a reproducible framework for this check, applicable to any TCGA cohort with partial clinical response data alongside survival annotations.

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Architecture

The software is organised into eight source modules under `src/`: `data_loader` (I/O), `preprocessing` (survival extraction, platinum-patient identification, censoring-aware label assignment), `features` (train-only variance selection, z-score scaling), `models` (classifier factories, CV utilities), `evaluation` (metrics, results tables), `visualization` (seven figure types), `ablation` (leakage conditions, bootstrap CIs, nested CV, paired significance tests), and `label_investigation` (coverage, distribution, consistency, cross-tabulation, exact binomial recall CIs).

All parameters are controlled via `config/config.yaml` (gitignored; a committed template `config/config.template.yaml` is provided). Three CLI entry points with `--help` (`run.py`, `run_ablation_study.py`, `investigate_clinical_labels.py`) enable full pipeline reproducibility from a single command.

Testing

The package includes 52 unit tests across three modules. All tests use synthetic data and pass without the TCGA dataset on disk, enabling CI/CD integration. Three tests are regression tests for bugs that only surfaced on real TCGA data (NaN values in RSEM matrices, mixed-type gene name indices) but were not caught by earlier synthetic-data tests — documenting a real gap between synthetic and real-data coverage.

Results

On the platinum-treated TCGA-OV cohort (n=214/206/199 at 12/18/24m), Logistic Regression achieved the highest test-set AUC at the 24-month endpoint (mean 0.70 across 30 repeated correct splits). The ablation study confirmed that unsupervised variance selection introduces negligible leakage while supervised F-score selection inflates AUC by a mean of +0.22, with SVM reaching AUC 1.00 under leaky conditions versus 0.45 under correct conditions at the 12-month endpoint. The label investigation found that the OS-threshold proxy had 0/5 recall (95% CI: [0.000, 0.522]) on the 7 clinically confirmed Progressive Disease patients, consistent with the mechanistic argument that overall survival and first-line treatment response are genuinely different clinical constructs.

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